

Quiz Week 9

Groups



Group Homomorphisms

1. In each case, determine whether $\varphi : G_1 \rightarrow G_2$ is a homomorphism and determine if it is injective and/ or surjective. Support your answer.
 - (a) $G_1 = G_2 = \mathbb{Z}_5^*$, $\varphi(g) = g^2$
 - (b) $G_1 = (M_n(\mathbb{R}), +)$, $(G_2 = \mathbb{R}, +)$, $\varphi(A) = \text{tr}(A)$
 - (c) $G_1 = G_2 = \mathbb{Z}_7$, $\varphi(g) = 2g$
 - (d) $G_1 = G_2 = \mathbb{C}^*$, $\varphi(z) = \bar{z}$ is the complex conjugation

Solution: The correct answers as per in the Google form are as follows:

The map in (a) is a homomorphism; the map in (a) is neither a monomorphism nor epimorphism; the map in (b) is a homomorphism; the map in (b) is an epimorphism but not a monomorphism; the map in (c) is a homomorphism; the map in (c) is an isomorphism; the map in (d) is a homomorphism; the map in (d) is an isomorphism.

Below you could find arguments supporting these answers.

(a) Since 5 is prime, we have that $\mathbb{Z}_5^* = \mathbb{Z}_5 - \{0\}$. Notice that φ is not injective since

$$\varphi(\bar{4}) = \bar{4}^2 = \bar{16} = \bar{1} = \varphi(\bar{1}).$$

On the other hand, φ is not surjective because if it were surjective, it would imply the injectivity since $\mathbb{Z}_5^*$ is a finite set. To finish, φ is a homomorphism since

$$\varphi(gh) = (gh)^2 = (gh)(gh) = g(hg)h = g(gh)h = g^2h^2,$$

for all $g, h \in \mathbb{Z}_5^*$ because $\mathbb{Z}_5^*$ is abelian.

(b) The map $\varphi : M_n(\mathbb{R}) \rightarrow G_2 = \mathbb{R}$ given by $\varphi(A) = \text{tr}(A)$ is a homomorphism because $\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B)$, for all $A, B \in M_n(\mathbb{R})$. It is not a monomorphism because φ is not injective. For example: $\text{tr}(e_{11}) = \text{tr}(e_{12}) = 1$. Here e_{ij} denotes the matrix having all its entries equal zero except the one in position ij that is 1. But φ is surjective since $r = \varphi(re_{11})$ for all $r \in \mathbb{R}$. Thus φ is an epimorphism.

(c) We first prove that φ is a homomorphism: for $g, h \in \mathbb{Z}_7$ we have that

$$\varphi(\bar{g} + \bar{h}) = \varphi(\overline{g+h}) = \overline{2g+h} = \overline{2(g+h)} = \overline{2g+2h} = \overline{2g} + \overline{2h} = \varphi(\bar{g}) + \varphi(\bar{h}).$$

Moreover, φ is an isomorphism because, different elements have distinct images and $\text{im } \varphi = \mathbb{Z}_7$. In fact:

$$\begin{aligned}\varphi(\bar{0}) &= \bar{0}, \\ \varphi(\bar{1}) &= \bar{2}, \\ \varphi(\bar{2}) &= \bar{4}, \\ \varphi(\bar{3}) &= \bar{6}, \\ \varphi(\bar{4}) &= \bar{8} = \bar{1}, \\ \varphi(\bar{5}) &= \bar{3}, \\ \varphi(\bar{6}) &= \bar{5}.\end{aligned}$$

(d) The complex conjugation map is a homomorphism since $\overline{zw} = \bar{z}\bar{w}$ for all $z, w \in \mathbb{C}^*$. Moreover, it is injective because if $\bar{z} = \bar{w}$ we have that $\bar{\bar{z}} = \bar{\bar{w}}$, i. e., $z = w$. It is also surjective because given $z \in \mathbb{C}^*$ we have that $z = \varphi(\bar{z})$.

2. Are $\mathbb{Z}_{10}^*$ and $\mathbb{Z}_{12}^*$ isomorphic?

Solution: First of all, notice that

$$\mathbb{Z}_{10}^* = \{\bar{1}, \bar{3}, \bar{7}, \bar{9}\}, \quad \mathbb{Z}_{12}^* = \{\bar{1}, \bar{5}, \bar{7}, \bar{11}\}.$$

These two groups can not be isomorphic because $\mathbb{Z}_{10}^*$ is a cyclic group, while $\mathbb{Z}_{12}^*$ is not. More precisely, we have that

$$\mathbb{Z}_{10}^* = \langle \bar{3} \rangle, \quad \text{since} \quad \bar{3}^2 = \bar{9}, \quad \bar{3}^3 = \bar{27} = \bar{7}, \quad \bar{3}^4 = \bar{81} = \bar{1}, \quad \text{in } \mathbb{Z}_{10}.$$

On the other hand, $\mathbb{Z}_{12}^*$ is not cyclic since all its element have order 2:

$$\bar{5}^2 = \bar{25} = \bar{1}, \quad \bar{7}^2 = \bar{49} = \bar{1}, \quad \bar{11}^2 = \bar{121} = \bar{1}, \quad \text{in } \mathbb{Z}_{12}.$$